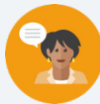


1.3 Isometry

Lesson Introduction

 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can define rigid motions. - I can show rigid motions by mapping using arrows on the plane. - I can explain why isometry preserves distance.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		N/A	
 Essential Questions	- When a shape is moved, does it change size? - What transformations result in a shape the remains the same size? - Can I identify the specific transformations that map a preimage to the image?			
 Lesson Narrative	This lesson allows students to engage with rigid transformations in a very kinesthetic and intentional way. Key vocabulary is defined that will be critical for learning throughout the remainder of the course. The primary learning outcome of this lesson is to identify transformations that do and do not preserve distance.			
 Component Summary	1.3.1 Warm-Up (~5 min.)	Notice and wonder of an image moving on pages (<i>SE p. 13</i>)	1.3.3 Bring It Together (10 min.)	Defining words associated with rigid motion (<i>SE p. 17</i>)
	1.3.2 Exploration (20 min.)	Exploring translations, reflections, and rotations with patty paper (<i>SE p. 14</i>)	1.3.4 Collaboration (10 min.)	Mapping rigid motion on gridded paper (<i>SE p. 18</i>)
	1.3.5 Wrap-Up (~5 min.)	Reflection on the lesson (<i>SE p. 19</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Reflection - Rotation - Translation - Isometry - Rigid Transformation - Rigid Motion - Congruent - Mapping - Line of Reflection Additional Terms: N/A		- Flipbook Video - Patty Paper - Ruler or Protractor (Optional) Sticky Notes	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions • Students may believe that if a shape looks the same, it is the same and measurement is not required to check. • Students may think diagonal movement is the same as a translation.	Things to Consider • For the transformations in 1.3.2 Exploration, consider asking students to label the vertices of each of the 5 images. • Consider a Foldable, Frayer Model, or type of interactive notebook for definitions throughout. • Consider allowing students to work through 1.3.2 Exploration and 1.3.4 Collaboration without worrying about having the “right” answers. By circulating and making notes about common errors and student misconceptions you can discuss these common errors and misconceptions in future lessons. Unit 1 is designed to help student explore some of the fundamental ideas of Geometry by utilizing transformations to have students experience them. The content repeatedly spirals back through the concepts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder For 1.3.1 Warm-Up, we use a Notice and Wonder to offer low-stakes questions inviting students to engage in the idea of rigid transformations. The animation presented is intended to create curiosity that could motivate more students to connect with the material in ways they previously would not have.	MA.K12.MTR.4.1: Discussion and Reflection Students will be implementing the Discussion and Reflection standard in 1.3.2 Exploration and 1.3.4 Collaboration. Students engage in discussions about mathematical thinking. Help to establish a culture where students ask questions to their peers and to you. Consider opportunities to present student work to help deepen student understanding.
 Differentiate Instruction	While developing a plan for helping students comprehend each of the six key words in 1.3.3 Bring It Together, consider having students develop a folded Frayer Model. This allows students to have a kinesthetic connection with these key words as they engage with each definition. Once completed, creating a pictorial representation of each word will also offer a visual entry-point for students.	
Lesson Notes:		

1.3.1 Warm-Up: Animation Flipbook

Component Overview: Play the Flipbook video for the entire class. After students watch the video (0:42 minutes), prompt them to reflect on the profession by answering the two questions independently.



1.3.1 Warm-Up: Animation Flipbook

An animation flipbook is a small book of drawings with a series of pictures that appear to move when the pages are flipped in rapid succession.



The video shows a series of pictures of two clouds. A few screenshots are shown.



1. What do you notice about the two clouds?
Answers will vary. The cloud on the right is happy, and the cloud on the left is sad.
2. At one point the two clouds seemed to move up and down. How do you think the artist was able to do that?
Answers will vary. The artist made small changes on each page to make it look like the clouds are moving when flipping through the book.



1.3.1 Warm-Up is designed to simply engage students. If students are missing the point or becoming confused, do not spend extra time clarifying here. This should be a quick, fun activity for students to become excited about the content. The video shows that the figures move but their size and shape do not change. Making this connection in 1.3.3 Bring It Together may be helpful to students. A further connection would be the application to video games where characters move but their size/shape do not change.



Should students become excited about this idea of a flipbook, it can make a great homework activity if it does not take much time away from this lesson. Consider prompting students to develop a flipbook that uses a key shape they learned from a previous class.

1.3.2 Exploration: Isometry

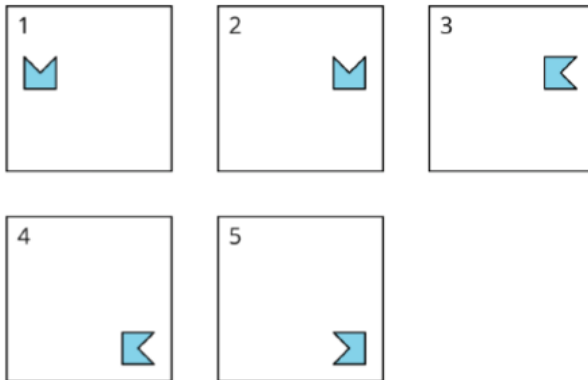
Component Overview: Students will explore moving the position of the image of a polygon using patty paper to determine if size and shape stays the same or changes for each type of movement.



1.3.2 Exploration: Isometry

Work with your partner to complete the following.

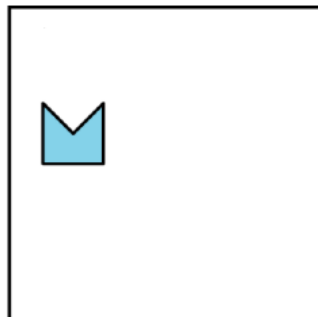
- Five frames that feature a polygon in different positions are shown.



Describe how the polygon moves from one frame to the next.

Movement from...	Description
Frame 1 to Frame 2	<i>Slide to the right</i>
Frame 2 to Frame 3	<i>Turn to the right</i>
Frame 3 to Frame 4	<i>Slide down</i>
Frame 4 to Frame 5	<i>Turn the other way</i>

- The diagram shows the position of the polygon from Frame 1.



This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 1.3.3 Bring It Together. Students may revise their thinking throughout the exploration as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 1.3.3 Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

Consider asking student to label corresponding vertices in each figure or color each corresponding edge the same color. Look for:

- Does the student translate or reflect from step 1 to step 2?
- Does the student translate or reflect from step 3 to step 4?
- Does the student reflect or rotate from step 4 to step 5?

1.3.2 Exploration: Isometry (continued)

- a. Label two pieces of patty paper as **A** and **B**. Then, complete the steps for each piece of patty paper.

Patty paper A

- Copy the polygon.
- Move patty paper **A** to display the polygon as it is shown in Frame 2.

Patty paper B

- Copy the polygon and the square that surrounds it.
- Fold the patty paper vertically so that the top corners of the square match up.
- With the patty paper folded, draw the copy of the polygon onto the other half of the patty paper.

- b. Complete the statements.

To draw the polygon as it is shown in Frame 2,

patty paper **A** was moved

- ☐ up.
- ☐ left.
- ☒ right.
- ☐ down.

When patty paper **B** was folded, the polygon could be traced so that the new polygon is as shown in Frame 2.

The original polygon was

- ☒ flipped across
- ☐ rotated around

the fold of the paper.

The fold of the paper was the

- ☐ center of rotation.
- ☒ line of reflection.



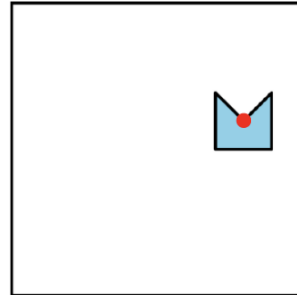
Students may wonder if the border of the square should be traced. It is not necessary for the border to be traced but a student may feel more comfortable doing so.



Scaffolding student responses help students to explain concepts they are just learning. Encourage students to read their completed sentence out loud to their partner. This will help students hear their own response and allow their partner to consider if they agree with the answer.

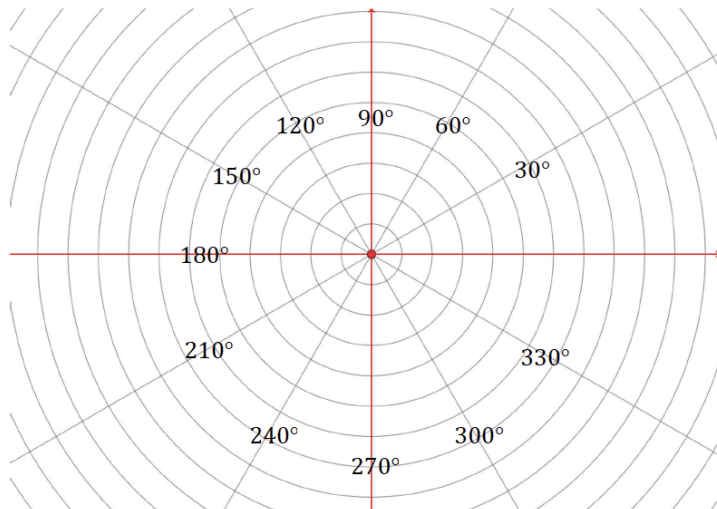
1.3.2 Exploration: Isometry (continued)

3. The diagram shows where the polygon is in Frame 2.



a. Complete the steps to find the point of rotation.

- Label a piece of patty paper as **C**.
- Copy the polygon onto patty paper **C**.
- Use the point of your writing utensil to find the point of rotation so the polygon can be rotated to look like Frame 3.
- Place a dot on the diagram above to show the point of rotation.
- Place patty paper **C** on the circular graph paper so the point of rotation is where the two red axes meet, and the polygon is oriented as in Frame 2.
- Rotate patty paper **C** so that it looks like Frame 3.



Students may benefit from a demonstration of how to use the point of their writing utensil to spin the patty paper around to find the point of rotation.

1.3.2 Exploration: Isometry (continued)

- b. Complete the statements.

To move the polygon so it looks like Frame 3, rotate the diagram on patty paper $\text{C } 90^\circ$ degrees clockwise $\cup$ or 270° degrees counterclockwise $\cap$.

The center of rotation is

- ☐ the top left vertex.
- ☐ the top right vertex.
- ☐ the bottom left vertex.
- ☐ the bottom right vertex.
- ☒ the center of the polygon.

4. Use patty paper, a ruler, or a protractor as necessary to complete the following.

- a. Complete the table by determining if the polygon changed in size or shape as it moved.

Movement from...	Did the size change?	Did the shape change?
Frame 1 to Frame 2	No	No
Frame 2 to Frame 3	No	No
Frame 3 to Frame 4	No	No
Frame 4 to Frame 5	No	No

- b. What tools or information were used when determining if the size changed?
Ruler and Patty Paper
- c. What tools or information were used when determining if the shape changed?
Protractor and Patty Paper
- d. Discuss with your partner if the measurement of angles in the polygon changed from Frame 1 to Frame 5.
No, they did not change.



Listen for students who think they can just “look” to determine if the size or shape changed. Help students understand the importance of precision.

1.3.3 Bring It Together: Definitions

Component Overview: Students can summarize or synthesize their findings in this activity. Translations capture horizontal and vertical transformations as seen in questions 1 and 2 in 1.3.2 Collaboration, while examples of rotations can be found in questions 3-5 of 1.3.2 Collaboration.



1.3.3 Bring It Together: Definitions

The movements of the polygon in the five frames are known as reflection, rotation, and translation. During the lesson a reflection was referred to as a flip, which is an informal definition. Another informal definition used was "is moved," which is formally known as a translation.



A **reflection** is a transformation that produces the mirror image of a geometric figure across a line of reflection.



A **rotation** is a transformation of a figure by turning it about a center point or axis. The amount of rotation can be expressed in the number of degrees. The direction of the rotation for two-dimensional figures can be expressed as clockwise or counterclockwise.



A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

Isometry is a transformation that preserves distance and angle measure.

1. Discuss with your partner the definition of isometry. Determine which transformations — reflection, rotation, or translation — preserve distance and angle measure. Summarize your discussion. *These three transformations preserve distance and angle measurements.*



A **rigid transformation** is a transformation of points in space consisting of a sequence of one or more translations, reflections, or rotations. Rigid transformations preserve distances and angle measures (congruency).



A rigid transformation is also called a **rigid motion**. Rigid means that something is not able to be changed. Isometry is composed of rigid motions.



Figures having exactly the same shape and size are **congruent**. Equivalently, two figures are congruent if one can be mapped to the other using a rigid transformation.



Consider a foldable, Frayer Model, or type of interactive notebook for these definitions. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

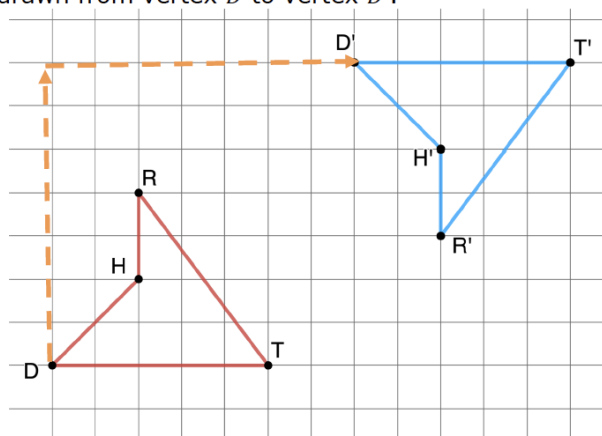
1.3.4 Collaboration: Isometry

Component Overview: Students map translations and rotations on gridded paper to determine that these transformations map all points on the figure.



1.3.4 Collaboration: Isometry

1. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to vertex D' .



$R'T'D'H'$ is read as "r prime, t prime, d prime, h prime". This notation is often used when a figure has been transformed.

The arrows show a possible path of movement. For the movement to be correct, the same path should exist for all points on a diagram.

Discuss with your partner if all of the points on $RTDH$ can be moved using the same path as point D .



A **mapping** is a standardized form of showing a transformation.



This is students' first introduction to mapping. Mapping shows the movement of a transformation. Mapping can be done with horizontal and vertical vectors or with a vector that is the hypotenuse of the triangle formed by the horizontal and vertical vectors. In preparation for students to describe a transformation algebraically using horizontal and vertical vectors may be easier for students. Algebraic descriptions using coordinate notation such as $(x, y) \rightarrow (x + h, y + k)$ are introduced in Unit 4. It is not necessary to use this notation in this unit but if your students are ready for this notation exposing them to the idea may lay a foundation for later units.

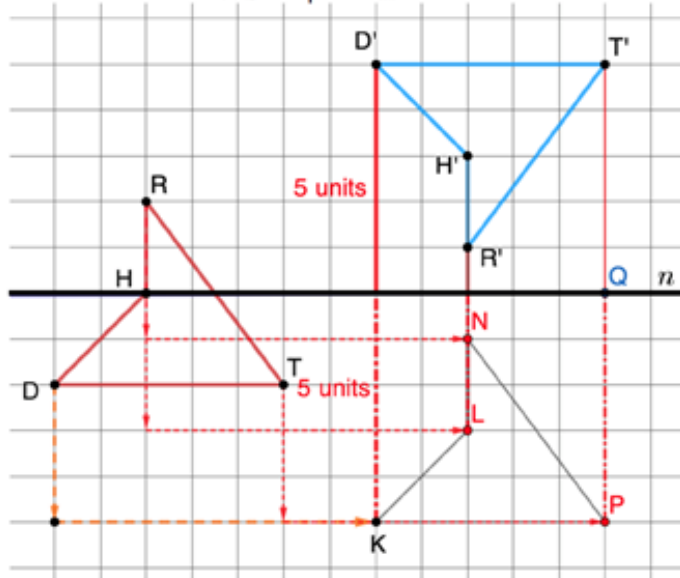
Also, note using notation, such as $(x, y) \rightarrow (x + h, y + k)$ can be called many things such as: algebraic descriptions, coordinate notation, or algebraic descriptions using coordinate notation. Utilize all of these names when discussing this representation.



This is a discussion opportunity. Students do not need to write an explanation here. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

1.3.4 Collaboration: Isometry (continued)

2. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to point K .



- Draw a line segment from point K to line n . What is the length of the line segment?
Shown on graph.
- Draw a line segment from point D' to line n . What is the length of the line segment?
Shown on graph.
- Describe the relationship each line segment has to line n .
Both line segments are perpendicular to line n .



If corresponding points have the same distance from a line then the line is the "**line of reflection**."



Students may make errors during 1.3.4 Collaboration, including mapping incorrectly. Allow the opportunity for them to correct their own thinking as they work through with their partner. Part of what students may realize by the end of the 1.3.4 Collaboration is that mapping one figure to another may be shown using the vertices but all of the points of the figure follow that same mapping. This idea is continually developed in Unit 1. Do not be afraid to let students make mistakes. As you circulate keep track of common errors by students that you can address in later lessons.



If students wonder it may be helpful to specify that the line segment should be the shortest distance from the point to the line, also known as the perpendicular distance.

1.3.4 Collaboration: Isometry (continued)

- d. For each of the other vertices on $RTDH$, draw a new point using the same movements done to map point D to point K . Then show the reflection to $R'T'D'H'$ using line segments.

Shown on graph.

- e. Discuss with your partner if the mapping of **all** of the points on $RTDH$ can be done using the same path and reflection, or if the mapping only works for the vertices. Summarize your discussion.

Answers will vary. Yes, all points can be mapped this way.



The focus of this component is to illustrate that “same size, same shape” is an inefficient definition of congruence. The foundation for the definition of congruence is conceptually taught early so that it can be applied throughout the course.

1.3.5 Wrap-Up: Reflection

Component Overview: Students respond to prompts of what they learned and what they still are wondering about from the lesson.

1.3.5 Wrap-Up: Reflection

- 1. Complete the table.
Answers will vary.

The terms and information used in the exploration that I knew were ...	Something I wondered during the exploration was ...	Something I learned during the exploration was ...



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons. Consider writing each student a brief note about their response or to have students anonymously share one of these reflections on a sticky note where you can write a positive note. Then, display all of the notes for students to view.